

实验六 变异鉴定

刘沛雨 2100012289 信息科学技术学院

版本信息

1. samtools: 1.10
2. bcftools: 1.12-51-g1b7cb73

练习 1：用 bwa 对大肠杆菌基因组测序样本进行序列比对

(1) 制作基因组索引（计 1 分）

```
[teach80_pkuhpc@login66 project]# ./data/bwa index -p step1_indexed_E.coli ./data/E.coli.fa
[bwa_index] Pack FASTA... 0.03 sec
[bwa_index] Construct BWT for the packed sequence...
[bwa_index] 0.78 seconds elapsed
[bwa_index] Update BWT... 0.02 sec
[bwa_index] Pack forward-only FASTA... 0.02 sec
[bwa_index] Construct SA from BWT and Occ... 0.43 sec
[main] Version: 0.7.17-r1188
[main] CMD: ./data/bwa index -p step1_indexed_E.coli ./data/E.coli.fa
[main] Real time: 1.533 sec; CPU: 1.280 sec
[teach80_pkuhpc@login66 project]# ls -l ./data/
total 30271586
-r--r--r-- 1 teach80_pkuhpc teach80_pkuhpc 4390912 Apr 1 13:40 E.coli.fa
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 199084896 Apr 1 13:40 SRR1770413
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 435681114 Apr 1 13:40 SRR1770413.2.fastq
lrwxrwxrwx 1 teach80_pkuhpc teach80_pkuhpc 46 Apr 1 13:40 bcftools -> /gpfs1/share/class6/software/bcftools/bcftools
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 29 Apr 1 13:40 E.coli.fa fai
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 26761350443 Apr 1 13:44 9086c9b5f797c70804060137f4b818525_gdc_realn.bam
-r--r--r-- 1 teach80_pkuhpc teach80_pkuhpc 3152288848 Apr 1 13:44 GRCh38.d1.vd1.fa
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 1755170 Apr 1 13:44 s34e77c6-g8c5-44a6-806f-fd4344284b19.vcf.gz
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 7647744 Apr 1 13:45 9086c9b5f797c70804060137f4b818525_gdc_realn.bam.bai
lrwxrwxrwx 1 teach80_pkuhpc teach80_pkuhpc 31 Apr 1 13:45 bwa -> /appsnew/bioapps/bwa-0.7.17/bwa
lrwxrwxrwx 1 teach80_pkuhpc teach80_pkuhpc 43 Apr 1 13:45 samtools -> /appsnew/bioapps/samtools-1.10/bin/samtools
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 129640 Apr 1 13:45 GRCh38.d1.vd1.fa fai
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 435681114 Apr 1 13:52 SRR1770413.1.fastq
[teach80_pkuhpc@login66 project]# ls -l ./data/
total 7490
drwxrwxr-x 2 teach80_pkuhpc teach80_pkuhpc 4096 Apr 1 13:52 data
drwxrwxr-x 2 teach80_pkuhpc teach80_pkuhpc 4096 Apr 1 13:53 bwa_results
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 4336712 Apr 1 13:54 step1_indexed_E.coli.bwt
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 1084160 Apr 1 13:54 step1_indexed_E.coli.pac
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 98 Apr 1 13:54 step1_indexed_E.coli.ann
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 12 Apr 1 13:54 step1_indexed_E.coli.amb
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 2168368 Apr 1 13:54 step1_indexed_E.coli.sa
[teach80_pkuhpc@login66 project]# mv ./step1_indexed_E.coli.* ./bwa_results/
[teach80_pkuhpc@login66 project]# ls -l ./bwa_results/
total 7489
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 4336712 Apr 1 13:54 step1_indexed_E.coli.bwt
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 1084160 Apr 1 13:54 step1_indexed_E.coli.pac
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 98 Apr 1 13:54 step1_indexed_E.coli.ann
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 12 Apr 1 13:54 step1_indexed_E.coli.amb
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 2168368 Apr 1 13:54 step1_indexed_E.coli.sa
[teach80_pkuhpc@login66 project]#
```

(2) 序列比对（计 1 分）

```
[teach80_pkuhpc@login66 bwa_results]# ./data/bwa mem -t 10 step1_indexed_E.coli ./data/SRR1770413.1.fastq ./data/SRR1770413.2.fastq > ./output.sam
[M::bwa_idx_load_from_disk] read 0 ALT contigs
[M::process] read 332226 sequences (100000026 bp)...
[M::mem_pestat] # candidate unique pairs for (FF, FR, RF, RR): (2, 106238, 2, 3)
[M::mem_pestat] skip orientation FF as there are not enough pairs
[M::mem_pestat] analyzing insert size distribution for orientation FR...
[M::mem_pestat] (25, 50, 75) percentile: (517, 580, 641)
[M::mem_pestat] low and high boundaries for computing mean and std.dev: (269, 889)
[M::mem_pestat] mean and std.dev: (576.98, 102.01)
[M::mem_pestat] low and high boundaries for proper pairs: (145, 1013)
[M::mem_pestat] skip orientation RF as there are not enough pairs
[M::mem_pestat] skip orientation RR as there are not enough pairs
[M::mem_process_seqs] Processed 332226 reads in 51.645 CPU sec, 5.200 real sec
[M::process] read 332226 sequences (100000026 bp)...
[M::mem_pestat] # candidate unique pairs for (FF, FR, RF, RR): (8, 106391, 0, 5)
[M::mem_pestat] skip orientation FF as there are not enough pairs
[M::mem_pestat] analyzing insert size distribution for orientation FR...
[M::mem_pestat] (25, 50, 75) percentile: (517, 580, 641)
[M::mem_pestat] low and high boundaries for computing mean and std.dev: (269, 889)
[M::mem_pestat] mean and std.dev: (576.25, 102.40)
[M::mem_pestat] low and high boundaries for proper pairs: (145, 1013)
[M::mem_pestat] skip orientation RF as there are not enough pairs
[M::mem_pestat] skip orientation RR as there are not enough pairs
[M::mem_process_seqs] Processed 332226 reads in 52.014 CPU sec, 5.241 real sec
[M::process] read 289028 sequences (87308228 bp)...
[M::mem_pestat] # candidate unique pairs for (FF, FR, RF, RR): (4, 107600, 1, 5)
[M::mem_pestat] skip orientation FF as there are not enough pairs
[M::mem_pestat] analyzing insert size distribution for orientation FR...
[M::mem_pestat] (25, 50, 75) percentile: (518, 581, 642)
[M::mem_pestat] low and high boundaries for computing mean and std.dev: (270, 890)
[M::mem_pestat] mean and std.dev: (577.36, 102.61)
[M::mem_pestat] low and high boundaries for proper pairs: (146, 1014)
[M::mem_pestat] skip orientation RF as there are not enough pairs
[M::mem_pestat] skip orientation RR as there are not enough pairs
[M::mem_process_seqs] Processed 332226 reads in 52.457 CPU sec, 5.352 real sec
[M::mem_pestat] # candidate unique pairs for (FF, FR, RF, RR): (4, 94088, 1, 1)
[M::mem_pestat] skip orientation FF as there are not enough pairs
[M::mem_pestat] analyzing insert size distribution for orientation FR...
[M::mem_pestat] (25, 50, 75) percentile: (510, 581, 642)
[M::mem_pestat] low and high boundaries for computing mean and std.dev: (270, 890)
[M::mem_pestat] mean and std.dev: (577.42, 102.23)
[M::mem_pestat] low and high boundaries for proper pairs: (146, 1014)
[M::mem_pestat] skip orientation RF as there are not enough pairs
[M::mem_pestat] skip orientation RR as there are not enough pairs
[M::mem_process_seqs] Processed 289828 reads in 45.272 CPU sec, 4.591 real sec
[main] Version: 0.7.17-r1188
[main] CMD: ./data/bwa mem -t 10 step1_indexed_E.coli ./data/SRR1770413.1.fastq ./data/SRR1770413.2.fastq
[main] Real time: 25.736 sec; CPU: 202.170 sec
[teach80_pkuhpc@login66 bwa_results]# ls -l ./
total 884769
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 4336712 Apr 1 13:54 step1_indexed_E.coli.bwt
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 1084160 Apr 1 13:54 step1_indexed_E.coli.pac
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 98 Apr 1 13:54 step1_indexed_E.coli.ann
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 12 Apr 1 13:54 step1_indexed_E.coli.amb
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 2168368 Apr 1 13:54 step1_indexed_E.coli.sa
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 898319428 Apr 1 14:06 output.sam
```

练习 2：将大肠杆菌基因组测序的 sam 文件转为 bam 文件，并对 bam 文件排序和建立索引（计 1 分）

(1) 文件转换

```
[[teach80_pkuhpc@login66 bwa_results]# ../data/samtools view -bS output.sam -o output.bam
[[teach80_pkuhpc@login66 bwa_results]# ls -lrt
total 1158273
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 4336712 Apr 1 13:54 step1_indexed_E_coli.bwt
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 1084160 Apr 1 13:54 step1_indexed_E_coli.pac
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 98 Apr 1 13:54 step1_indexed_E_coli.ann
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 12 Apr 1 13:54 step1_indexed_E_coli.amb
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 2168368 Apr 1 13:54 step1_indexed_E_coli.sa
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 898319428 Apr 1 14:06 output.sam
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 280064621 Apr 1 14:28 output.bam
```

(2) 排序

```
[[teach80_pkuhpc@login66 bwa_results]# ../data/samtools sort output.bam -m 768M -o sorted_output.bam -@10
[bam_sort_core] merging from 0 files and 10 in-memory blocks...
[[teach80_pkuhpc@login66 bwa_results]# ls -lrt sorted_output.bam
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 207264265 Apr 1 14:31 sorted_output.bam
[[teach80_pkuhpc@login66 bwa_results]# ls -lrt
total 1360705
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 4336712 Apr 1 13:54 step1_indexed_E_coli.bwt
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 1084160 Apr 1 13:54 step1_indexed_E_coli.pac
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 98 Apr 1 13:54 step1_indexed_E_coli.ann
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 12 Apr 1 13:54 step1_indexed_E_coli.amb
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 2168368 Apr 1 13:54 step1_indexed_E_coli.sa
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 898319428 Apr 1 14:06 output.sam
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 280064621 Apr 1 14:28 output.bam
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 207264265 Apr 1 14:31 sorted_output.bam
[[teach80_pkuhpc@login66 bwa_results]#
```

(3) 索引

```
[[teach80_pkuhpc@login66 bwa_results]# ../data/samtools index sorted_output.bam -@ 10
[[teach80_pkuhpc@login66 bwa_results]# ls -lrt
total 1457089
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 4336712 Apr 1 13:54 step1_indexed_E_coli.bwt
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 1084160 Apr 1 13:54 step1_indexed_E_coli.pac
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 98 Apr 1 13:54 step1_indexed_E_coli.ann
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 12 Apr 1 13:54 step1_indexed_E_coli.amb
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 2168368 Apr 1 13:54 step1_indexed_E_coli.sa
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 898319428 Apr 1 14:06 output.sam
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 280064621 Apr 1 14:28 output.bam
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 207264265 Apr 1 14:31 sorted_output.bam
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 13616 Apr 1 14:36 indexed_sorted_output.bam.bai
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 98611763 Apr 1 14:59 sorted_output_depth.txt
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 13616 Apr 1 15:04 sorted_output.bam.bai
```

扩展：得到每个碱基位点的测序深度

```
[[teach80_pkuhpc@login66 bwa_results]# ../data/samtools depth -a sorted_output.bam > sorted_output_depth.txt
[[teach80_pkuhpc@login66 bwa_results]# ls -lrt
total 1457057
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 4336712 Apr 1 13:54 step1_indexed_E_coli.bwt
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 1084160 Apr 1 13:54 step1_indexed_E_coli.pac
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 98 Apr 1 13:54 step1_indexed_E_coli.ann
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 12 Apr 1 13:54 step1_indexed_E_coli.amb
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 2168368 Apr 1 13:54 step1_indexed_E_coli.sa
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 898319428 Apr 1 14:06 output.sam
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 280064621 Apr 1 14:28 output.bam
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 207264265 Apr 1 14:31 sorted_output.bam
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 13616 Apr 1 14:36 indexed_sorted_output.bam.bai
-rw-rw-r-- 1 teach80_pkuhpc teach80_pkuhpc 98611763 Apr 1 14:59 sorted_output_depth.txt
```

得到的部分测序深度如下所示：

NC_000913.3	1	46
NC_000913.3	2	47
NC_000913.3	3	49
NC_000913.3	4	51
NC_000913.3	5	51
NC_000913.3	6	51
NC_000913.3	7	51
NC_000913.3	8	51
NC_000913.3	9	52

练习 3: 用 samtools/bcftools 流程鉴定变异, 输出 vcf 文件 (计 2 分)

```
[teach80_kpuhpc@login66 bwa_results]$ ./data/bcftools mpileup -Ou -f ./data/E.coli.fa ./sorted_output.bam | ./data/bcftools call --ploidy 1 -mv > var.vcf
[mpileup] 1 samples in 1 input files
[mpileup] maximum number of reads per input file set to -d 250
[teach80_kpuhpc@login66 bwa_results]$ ls -lrt
total 1457121
-rw-rw-r-- 1 teach80_kpuhpc teach80_kpuhpc 4336712 Apr 1 13:54 step1_indexed_E.coli.bwt
-rw-rw-r-- 1 teach80_kpuhpc teach80_kpuhpc 1084160 Apr 1 13:54 step1_indexed_E.coli.pac
-rw-rw-r-- 1 teach80_kpuhpc teach80_kpuhpc 98 Apr 1 13:54 step1_indexed_E.coli.ann
-rw-rw-r-- 1 teach80_kpuhpc teach80_kpuhpc 12 Apr 1 13:54 step1_indexed_E.coli.amb
-rw-rw-r-- 1 teach80_kpuhpc teach80_kpuhpc 2168368 Apr 1 13:54 step1_indexed_E.coli.sa
-rw-rw-r-- 1 teach80_kpuhpc teach80_kpuhpc 898319428 Apr 1 14:06 output.sam
-rw-rw-r-- 1 teach80_kpuhpc teach80_kpuhpc 280064621 Apr 1 14:28 output.bam
-rw-rw-r-- 1 teach80_kpuhpc teach80_kpuhpc 207264265 Apr 1 14:31 sorted_output.bam
-rw-rw-r-- 1 teach80_kpuhpc teach80_kpuhpc 13616 Apr 1 14:36 indexed_sorted_output.bam.bai
-rw-rw-r-- 1 teach80_kpuhpc teach80_kpuhpc 98611763 Apr 1 14:59 sorted_output_depth.txt
-rw-rw-r-- 1 teach80_kpuhpc teach80_kpuhpc 13616 Apr 1 15:04 sorted_output.bam.bai
-rw-rw-r-- 1 teach80_kpuhpc teach80_kpuhpc 17984 Apr 1 15:14 var.vcf
[teach80_kpuhpc@login66 bwa_results]$
```

得到的 vcf 文件部分内容如下:

```

#filterformat=VCFv4.2
##FILTER=ID=PASS,Description="All filters passed"
##bcftoolsVersion=1.12-gb7cb73b1b12-32-b818088a
##bcftoolsCommandline=cp -Ou -f /data/E.coli.fa /sorted_output.bam
##reference=file:///data/E.coli.fa
##contig=ID=NC_008913.3,length=4336633
##ALT=ID=Desc,Description="Represents alleles (other than observed.)"
##INFO=ID=INDEL,Number=0,Type=Flag,Description="Indicates that the variant is an INDEL."
##INFO=ID=IDV,Number=1,Type=Integer,Description="Maximum number of raw reads supporting an indel"
##INFO=ID=IMF,Number=1,Type=Float,Description="Maximum fraction of raw reads supporting an indel"
##INFO=ID=IDP,Number=1,Type=Integer,Description="Raw read depth"
##INFO=ID=IDV,Number=1,Type=Float,Description="Variant Distance Bias for filtering splice-site artefacts in RNA-seq data (bigger is better)",Version="3"
##INFO=ID=IDRPZ,Number=1,Type=Float,Description="Mann-Whitney U-z test of Read Position Bias (closer to 0 is better)"
##INFO=ID=IDQB2,Number=1,Type=Float,Description="Mann-Whitney U-z test of Mapping Quality Bias (closer to 0 is better)"
##INFO=ID=IDQB2,Number=1,Type=Float,Description="Mann-Whitney U-z test of Base Quality Bias (closer to 0 is better)"
##INFO=ID=IDMSB2,Number=1,Type=Float,Description="Mann-Whitney U-z test of Mapping Quality vs Strand Bias (closer to 0 is better)"
##INFO=ID=IDSCB,Number=1,Type=Float,Description="Mann-Whitney U-z test of Soft-Clip Length Bias (closer to 0 is better)"
##INFO=ID=IDDS,Number=1,Type=Integer,Description="Phred-scaled p-value using Fisher's exact test to detect strand bias"
##INFO=ID=IDSB,Number=1,Type=Float,Description="Segregation based metric."
##INFO=ID=IDMQ,Number=1,Type=Float,Description="Fraction of MQ0 reads (smaller is better)"
##FORMAT=ID=PL,Number,Description="List of Phred-scaled genotype likelihoods"
##FORMAT=ID=GT,Number=1,Type=String,Description="Genotype"
##INFO=ID=IDAC,Number,A,Type=Integer,Description="Allele count in genotypes for each ALT allele, in the same order as listed"
##INFO=ID=IDAN,Number=1,Type=Integer,Description="Total number of alleles in called genotypes"
##INFO=ID=IDPA,Number=A,Type=Integer,Description="Number of high-quality reads ref-forward, ref-reverse, alt-forward and alt-reverse bases"
##INFO=ID=IDMQ,Number=1,Type=Integer,Description="Average mapping quality"
##bcftools_callVersion=1.12-gb7cb73b1b12-32-b818088a
##bcftools_callCommandline=cp -ploidy 1 - 15:13:34.2025
#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT /sorted_output.bam
NC_00913.3 29568 . C T 21.4455 . DP=2;VDB=0.02;SDB=0.453602;FS=0;MQ0F=0;AC=1;AN=1;DP4=0.0,2.0;MQ=27 GT:PL 1:51,0
NC_00913.3 29571 . T C 21.4455 . DP=2;VDB=0.02;SDB=0.453602;FS=0;MQ0F=0;AC=1;AN=1;DP4=0.0,2.0;MQ=27 GT:PL 1:51,0
NC_00913.3 29577 . G C 21.4455 . DP=2;VDB=0.02;SDB=0.453602;FS=0;MQ0F=0;AC=1;AN=1;DP4=0.0,2.0;MQ=27 GT:PL 1:51,0
NC_00913.3 29586 . G T 21.4455 . DP=2;VDB=0.02;SDB=0.453602;FS=0;MQ0F=0;AC=1;AN=1;DP4=0.0,2.0;MQ=27 GT:PL 1:51,0
NC_00913.3 29589 . T C 21.4455 . DP=2;VDB=0.02;SDB=0.453602;FS=0;MQ0F=0;AC=1;AN=1;DP4=0.0,2.0;MQ=27 GT:PL 1:51,0
NC_00913.3 29592 . T C 21.4455 . DP=2;VDB=0.02;SDB=0.453602;FS=0;MQ0F=0;AC=1;AN=1;DP4=0.0,2.0;MQ=27 GT:PL 1:51,0
NC_00913.3 29610 . G A 46.416 . DP=3;VDB=0.490895;SDB=0.511535;FS=0;MQ0F=0;AC=1;AN=1;DP4=0.0,3.0;MQ=34 GT:PL 1:76,0
NC_00913.3 29673 . C A 158.416 . DP=12;VDB=0.111776;SDB=0.688642;FS=0;MQ0F=0;AC=1;AN=1;DP4=0.0,12.0;MQ=54 GT:PL 1:181,0
NC_00913.3 29679 . C A 151.416 . DP=12;VDB=0.44086.85;SDB=0.688642;FS=0;MQ0F=0;AC=1;AN=1;DP4=0.0,12.0;MQ=54 GT:PL 1:180,0
NC_00913.3 29681 . C A 158.416 . DP=13;VDB=0.89581.85;SDB=0.68931.85;MQ0F=0;AC=1;AN=1;DP4=0.0,13.0;MQ=54 GT:PL 1:178,0
NC_00913.3 296784 . C G 159.416 . DP=15;VDB=0.617576;SDB=0.688148;FS=0;MQ0F=0;AC=1;AN=1;DP4=0.0,15.0;MQ=54 GT:PL 1:189,0
NC_00913.3 296793 . G A 159.416 . DP=15;VDB=0.00817474;SDB=0.688148;FS=0;MQ0F=0;AC=1;AN=1;DP4=0.0,15.0;MQ=54 GT:PL 1:189,0
NC_00913.3 296988 . G A 164.41 . DP=16;VDB=0.181226.85;SDB=0.688148;RPBZ=0.165647;MQBZ=1.45118;MQSBZ=1.5983;SCBZ=1.52187;FS=0;MQ0F=0;AC=1;AN=1;DP4=0.0,15.0;MQ=53 GT:PL 1:191,0
0
NC_00913.3 296814 . A G 161.416 . DP=15;VDB=0.015088.85;SDB=0.688148;FS=0;MQ0F=0;AC=1;AN=1;DP4=0.0,15.0;MQ=53 GT:PL 1:191,0
NC_00913.3 296817 . G A 161.416 . DP=15;VDB=0.002756.85;SDB=0.688148;FS=0;MQ0F=0;AC=1;AN=1;DP4=0.0,15.0;MQ=53 GT:PL 1:191,0
NC_00913.3 296820 . T G 163.416 . DP=15;VDB=0.67248.85;SDB=0.688148;FS=0;MQ0F=0;AC=1;AN=1;DP4=0.0,15.0;MQ=53 GT:PL 1:188,0
NC_00913.3 296823 . G A 158.421 . DP=15;VDB=0.481836.85;SDB=0.688148;FS=0;MQ0F=0;AC=1;AN=1;DP4=0.0,15.0;MQ=53 GT:PL 1:191,0
NC_00913.3 296838 . C T 166.416 . DP=18;VDB=0.0104837;SDB=0.691153;FS=0;MQ0F=0;AC=1;AN=1;DP4=0.0,18.0;MQ=52 GT:PL 1:196,0
NC_00913.3 296841 . G C 163.416 . DP=18;VDB=0.0104837;SDB=0.691153;FS=0;MQ0F=0;AC=1;AN=1;DP4=0.0,18.0;MQ=52 GT:PL 1:196,0
NC_00913.3 296850 . C T 166.416 . DP=18;VDB=0.0154543;SDB=0.691153;FS=0;MQ0F=0;AC=1;AN=1;DP4=0.0,18.0;MQ=52 GT:PL 1:196,0
NC_00913.3 296856 . C G 164.416 . DP=17;VDB=0.00832848;SDB=0.690438;FS=0;MQ0F=0;AC=1;AN=1;DP4=0.0,17.0;MQ=54 GT:PL 1:194,0
NC_00913.3 296861 . G A 167.416 . DP=17;VDB=0.0138989;SDB=0.690438;FS=0;MQ0F=0;AC=1;AN=1;DP4=0.0,17.0;MQ=54 GT:PL 1:197,0
NC_00913.3 296863 . G C 177.416 . DP=18;VDB=0.068137;SDB=0.691153;MQSBZ=1.47799;FS=0;MQ0F=0;AC=1;AN=1;DP4=0.0,17.1;MQ=54 GT:PL 1:207,0
NC_00913.3 296877 . C G 174.416 . DP=17;VDB=0.0787409;SDB=0.690438;MQSBZ=1.46618;FS=0;MQ0F=0;AC=1;AN=1;DP4=0.0,16.1;MQ=54 GT:PL 1:204,0
NC_00913.3 296879 . T G 175.416 . DP=17;VDB=0.096112;SDB=0.690438;MQSBZ=1.46618;FS=0;MQ0F=0;AC=1;AN=1;DP4=0.0,16.1;MQ=54 GT:PL 1:205,0
NC_00913.3 296881 . G C 175.416 . DP=17;VDB=0.09227;SDB=0.690438;MQSBZ=1.46618;FS=0;MQ0F=0;AC=1;AN=1;DP4=0.0,16.1;MQ=54 GT:PL 1:205,0
NC_00913.3 296883 . G A 152.39 . DP=17;VDB=0.147615;SDB=0.689464;RPBZ=0.821964;MQBZ=0.523635;MQSBZ=1.46618;MQSBZ=1.1895;SCBZ=0.694138;FS=0;MQ0F=0;AC=1;AN=1;DP4=0.0,15.1;MQ=54 GT:PL 1:192,0
11379.0
NC_00913.3 296898 . G C 196.416 . DP=18;VDB=0.0565897;SDB=0.691153;MQSBZ=1.3832;FS=0;MQ0F=0;AC=1;AN=1;DP4=0.0,17.1;MQ=55 GT:PL 1:226,0
NC_00913.3 296901 . C G 181.416 . DP=18;VDB=0.5892;SDB=0.691153;MQSBZ=1.3832;FS=0;MQ0F=0;AC=1;AN=1;DP4=0.0,17.1;MQ=55 GT:PL 1:211,0
NC_00913.3 296904 . A C 178.416 . DP=18;VDB=0.639832;SDB=0.691153;MQSBZ=1.3832;FS=0;MQ0F=0;AC=1;AN=1;DP4=0.0,17.1;MQ=55 GT:PL 1:208,0
NC_
```

扩展：使用 samtools tview 查看某处变异

```
[teach80_pkuhpc@login66 bwa_results]$ ../data/samtools tview --reference ../data/E.coli.fa ./sorted_output.bam
[teach80_pkuhpc@login66 bwa_results]$
```

289571 289581 289591 289601 289611 289621
 CTCTGCCCACTTCTCTTTGGCTCGGCCATCGACACCCACAGTCGGTAGATAAAAGT
 T..C..T...T..T..C..C...CGAC...A...A...
 T..C...T...T...T..C...A...
 T..C...T...T...T..C...C..A...
A..

这里查看了 NC_000913.3:289568 – NC_000913.3:289621，结果与 `vcf` 文件中显示的变异位点与变异类型一致。

注：未使用 AI 协助完成实验或报告。